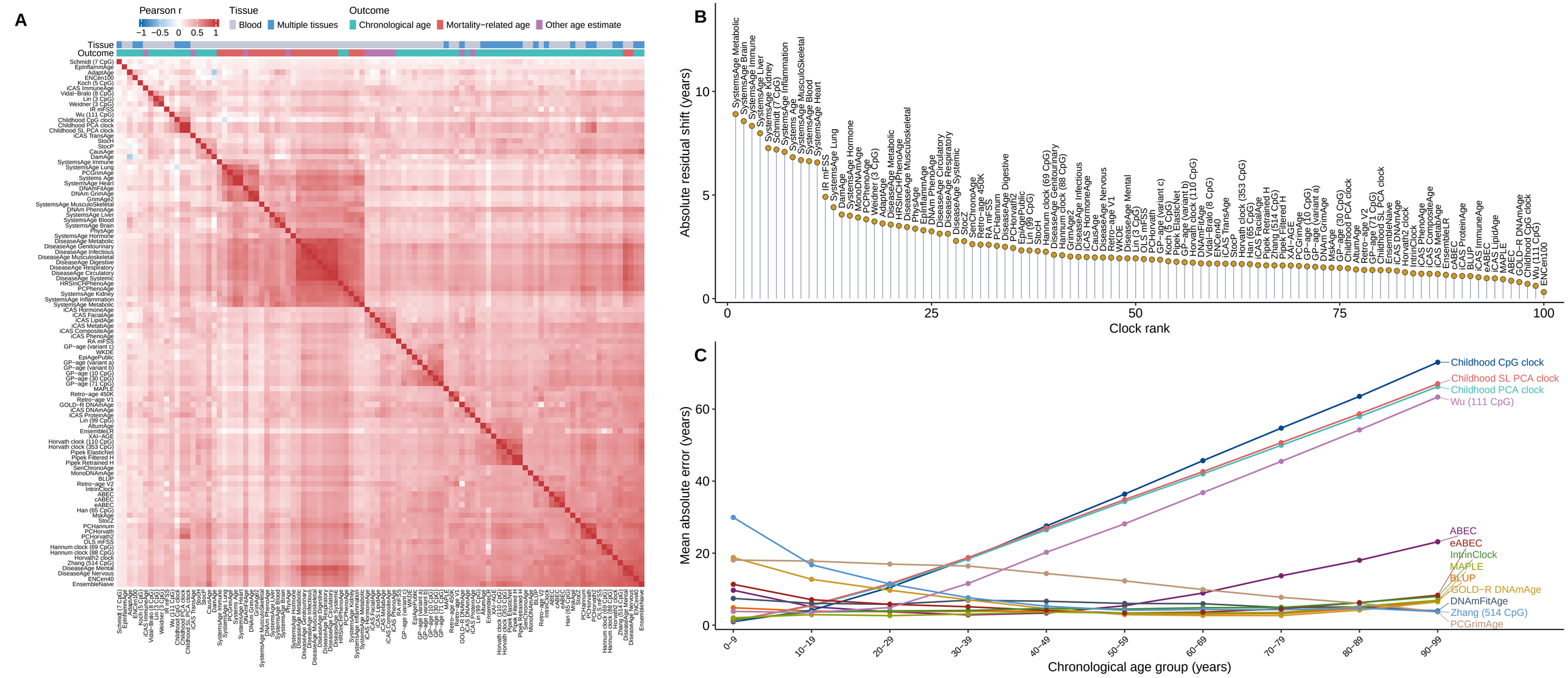
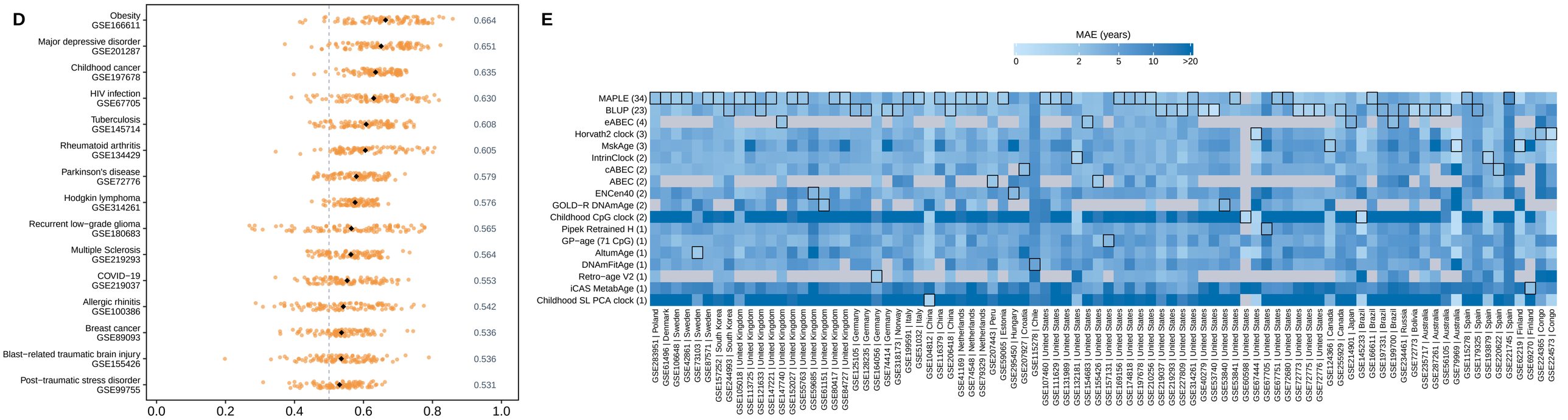
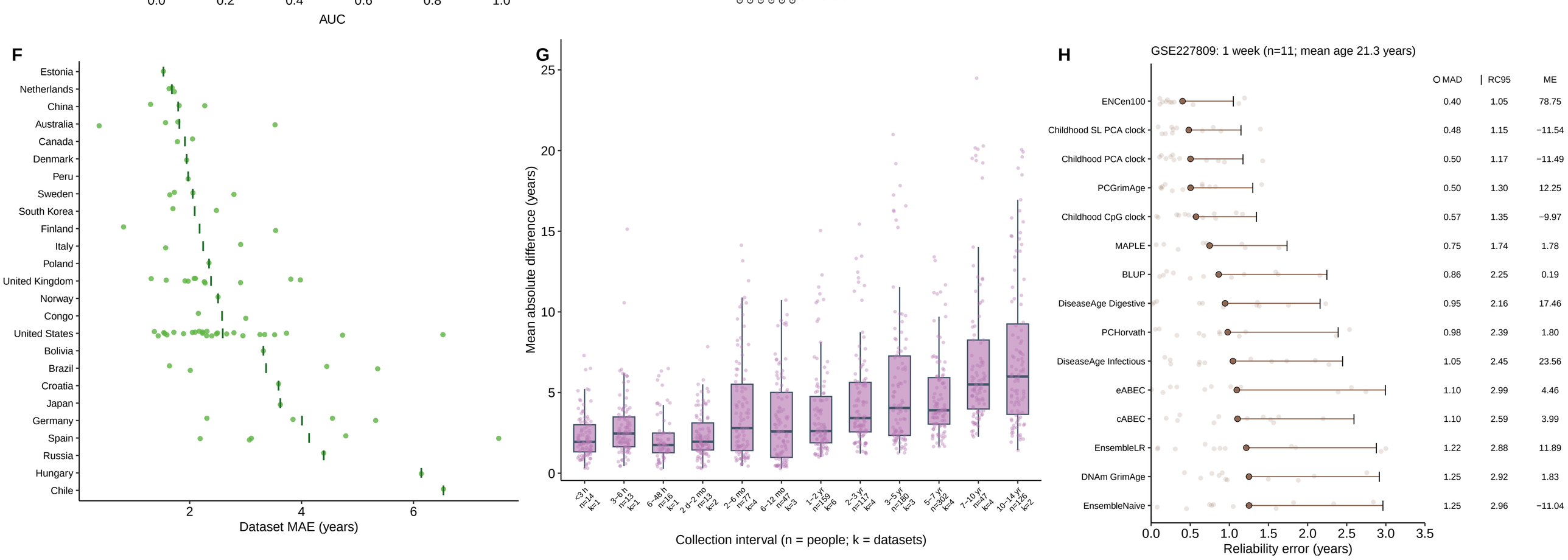
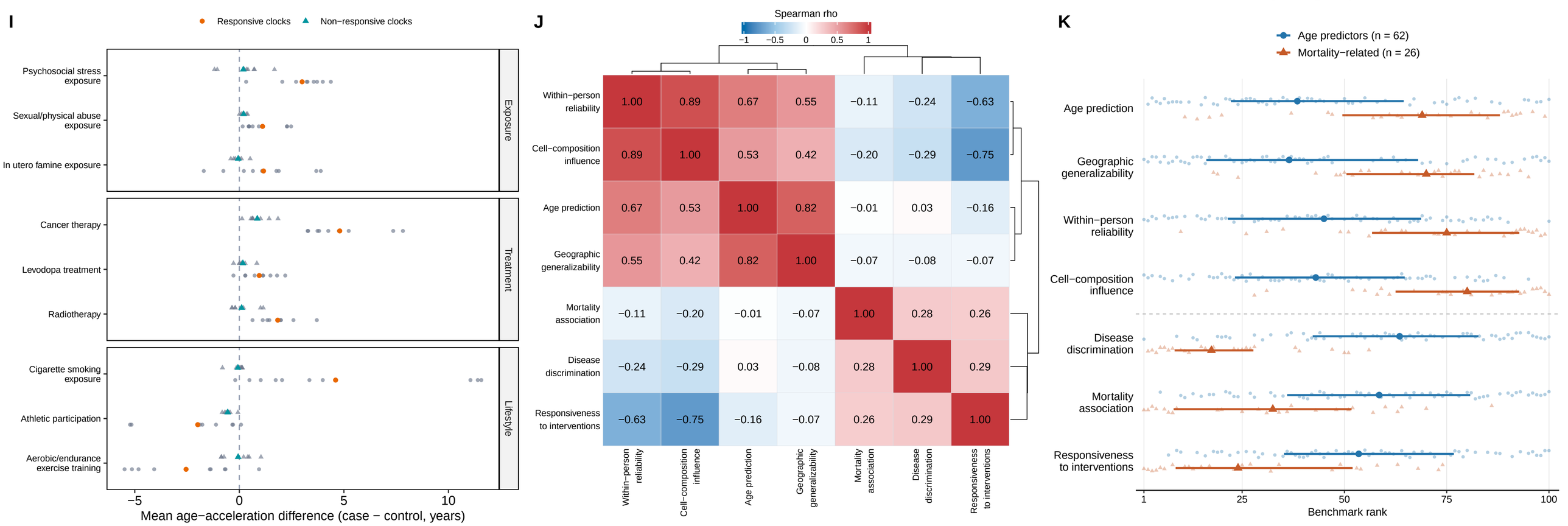

A
Pearson r
−1 −0.5 0 0.5 1
Tissue
Blood
Multiple tissues
Outcome
Chronological age
Mortality−related age
Other age estimate
B
Absolute residual shift (years)
Clock rank
C
Mean absolute error (years)
Childhood CpG clock
Childhood SL PCA clock
Childhood PCA clock
Wu (111 CpG)
ABEC
eABEC
IntrinClock
MAPLE
BLUP
GOLD−R DNAmAge
DNAmFitAge
Zhang (514 CpG)
PCGrimAge
Chronological age group (years)


D
Obesity GSE166611 0.664
Major depressive disorder GSE201287 0.651
Childhood cancer GSE197678 0.635
HIV infection GSE67705 0.630
Tuberculosis GSE145714 0.608
Rheumatoid arthritis GSE134429 0.605
Parkinson's disease GSE72776 0.579
Hodgkin lymphoma GSE314261 0.576
Recurrent low−grade glioma GSE180683 0.565
Multiple Sclerosis GSE219293 0.564
COVID−19 GSE219037 0.553
Allergic rhinitis GSE100386 0.542
Breast cancer GSE89093 0.536
Blast−related traumatic brain injury GSE155426 0.536
Post−traumatic stress disorder GSE99755 0.531
AUC
E
MAE (years)
0 2 5 10 >20
MAPLE (34)
BLUP (23)
eABEC (4)
Horvath2 clock (3)
MskAge (3)
IntrinClock (2)
cABEC (2)
ABEC (2)
ENCen40 (2)
GOLD−R DNAmAge (2)
Childhood CpG clock (2)
Pipek Retrained H (1)
GP−age (71 CpG) (1)
AltumAge (1)
DNAmFitAge (1)
Retro−age V2 (1)
iCAS MetabAge (1)
Childhood SL PCA clock (1)


F
Estonia
Netherlands
China
Australia
Canada
Denmark
Peru
Sweden
South Korea
Finland
Italy
Poland
United Kingdom
Norway
Congo
United States
Bolivia
Brazil
Croatia
Japan
Germany
Spain
Russia
Hungary
Chile
Dataset MAE (years)
G
Mean absolute difference (years)
Collection interval (n = people; k = datasets)
H
GSE227809: 1 week (n=11; mean age 21.3 years)
MAD
RC95
ME
ENCen100 0.40 1.05 78.75
Childhood SL PCA clock 0.48 1.15 −11.54
Childhood PCA clock 0.50 1.17 −11.49
PCGrimAge 0.50 1.30 12.25
Childhood CpG clock 0.57 1.35 −9.97
MAPLE 0.75 1.74 1.78
BLUP 0.86 2.25 0.19
DiseaseAge Digestive 0.95 2.16 17.46
PCHorvath 0.98 2.39 1.80
DiseaseAge Infectious 1.05 2.45 23.56
eABEC 1.10 2.99 4.46
cABEC 1.10 2.59 3.99
EnsembleLR 1.22 2.88 11.89
DNAm GrimAge 1.25 2.92 1.83
EnsembleNaive 1.25 2.96 −11.04
Reliability error (years)


I
Responsive clocks
Non−responsive clocks
Psychosocial stress exposure
Sexual/physical abuse exposure
In utero famine exposure
Exposure
Cancer therapy
Levodopa treatment
Radiotherapy
Treatment
Cigarette smoking exposure
Athletic participation
Aerobic/endurance exercise training
Lifestyle
Mean age−acceleration difference (case − control, years)
J
Spearman rho
−1 −0.5 0 0.5 1
Within−person reliability 1.00 0.89 0.67 0.55 −0.11 −0.24 −0.63
Cell−composition influence 0.89 1.00 0.53 0.42 −0.20 −0.29 −0.75
Age prediction 0.67 0.53 1.00 0.82 −0.01 0.03 −0.16
Geographic generalizability 0.55 0.42 0.82 1.00 −0.07 −0.08 −0.07
Mortality association −0.11 −0.20 −0.01 −0.07 1.00 0.28 0.26
Disease discrimination −0.24 −0.29 0.03 −0.08 0.28 1.00 0.29
Responsiveness to interventions −0.63 −0.75 −0.16 −0.07 0.26 0.29 1.00
K
Age predictors (n = 62)
Mortality−related (n = 26)
Age prediction
Geographic generalizability
Within−person reliability
Cell−composition influence
Disease discrimination
Mortality association
Responsiveness to interventions
Benchmark rank